

Tristan Lecourtois

Education

- 2025–2026 **Master of Science in Mathematics, Vision, Learning (MVA)**
École Normale Supérieure Paris-Saclay
Machine Learning, Deep Learning, Optimization, Reinforcement Learning, NLP
- 2023–2026 **Master of Engineering**
Mines de Saint-Étienne, France
Computer Science (Machine Learning, Python, C, C++, Algorithmics, SQL)
- 2021–2023 **Classes Préparatoires (MPSI/MP)**
Lycée Pierre Corneille, Rouen
Equivalent to a Bachelor's degree in Mathematics and Physics

Research Experience

- Mar. 2026 – June 2026 **Research Intern.**
InstaDeep, Paris
Pre-trained a multimodal foundation model from scratch in oncology (H&E histopathology, RNA-seq, DNA) via large-scale self-supervised masked modeling with distributed multi-GPU training. Built evaluation and benchmarking pipelines against state-of-the-art foundation-model baselines on downstream clinical tasks.
- Apr. 2025 – Aug. 2025 **Visiting Student Researcher.**
NASA Jet Propulsion Laboratory (JPL), Los Angeles, CA
Extended [ROSA](#), an LLM/VLM-powered agent for natural language control of ROS-based robots via a reasoning-action-observation loop. Prototyped episodic memory for contextual recall, and a DINOv2 + SAM vision pipeline for grounding-based object detection and segmentation.
- Jan. 2024 – June 2024 **Research Intern.**
Atos, Aix-en-Provence
Fine-tuned open-source foundation models (LLMs) for Text-to-SQL, reaching 90% query execution accuracy. Built a Retrieval-Augmented Generation (RAG) framework with LangChain and Ollama for conversational access to structured and unstructured data.

Research Projects

- June 2026 **World-Model Safety Watchdog for Robots** [\[Blog\]](#)
YC Paris × Unaiite Hackathon – 1st Global Prize
Built an external, embodiment-agnostic safety layer that predicts dangerous robot actions before they occur, coupling a VLA (semantic danger prior via structured outputs) with a JEPA world-model (LeWorldModel) whose latent-space prediction error yields a physical surprise/danger signal. Fused with an open-vocabulary perception stack (YOLOE + Segment Anything + MediaPipe).

- June 2026 **Drug-Perturbation JEPA World-Model** [\[Code\]](#) [\[Slides\]](#)
World Model Hackathon
 Built an energy-based JEPA world-model that predicts how a cell's transcriptomic state changes under a drug intervention, learning the perturbation entirely in latent space on the Tahoe-100M single-cell dataset. Extended eb-JEPA with a drug-conditioned predictor, anti-collapse regularization, and biology-informed losses, enabling in-silico drug screening.
- Dec. 2025 **Cross-Modal Alignment for Molecular Graph Captioning** [\[Code\]](#) [\[Report\]](#)
ALTeGraD Course, ENS Paris-Saclay
 Built a hybrid cross-modal framework (MolCA-based) translating molecular graphs into captions, aligning a GINE graph encoder with a Galactica-1.3B LLM via a Q-Former projector. Benchmarked three approaches: contrastive dual-encoder retrieval (SciBERT/RoBERTa), GNN-to-T5 generation, and soft-prompting.

Technical Strengths

- Languages Python, C++, LaTeX, HTML, CSS
 Frameworks PyTorch, TensorFlow, NumPy, OpenCV, Pandas, scikit-learn, ROS
 Tools Linux, Git, Docker

Relevant Coursework

- ENS Paris-Saclay Representation Learning for Computer Vision and Medical Imaging, Reinforcement Learning, Machine Learning for Time Series, Algorithms for speech and NLP, Geometric Data Analysis, Advanced Learning for Text and Graph Data, Generative Modeling for images

Awards and Achievements

- 2026 🏆 **1st Global Prize – YC Paris × UnaiHackathon “Paris Builds”** (sponsored by Y Combinator, Anthropic, QRT & Hugging Face) – Built Vigie, a world-model safety watchdog predicting dangerous robot actions before they happen (*Blog*)
- 2026 **Mistral AI Worldwide Hackathon (Top 100 / 7,000+ applicants)** – Developed AI-powered 911 emergency dispatch system (*Code & Demo*)
- 2026 **Participant at HackEurope, the largest student hackathon in Europe** – Developed Feelow, a personal finance Agent analyzing Polymarket prediction markets to generate trading insights, combining LLM-powered search, sentiment analysis, and quantitative scoring (*Code & Demo*)
- 2025 **Recipient of the Fondation Mines-Télécom International Scholarship** for research at NASA JPL
- 2025 **Participant at UC Berkeley AI Hackathon** – Developed an AI Therapist project (*Project Link*)
- 2025 **Y Combinator AI Startup School** – Selected participant for the intensive AI entrepreneurship program